Finite-Dimensional Competition and the Structural Necessity of Hierarchical Modularity

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Abstract

This work develops a structural account of biological evolution grounded in the principles of relation, coherence, and closure. When functional units compete within a finite-dimensional environmental constraint space, non-vanishing overlap of survival profiles becomes unavoidable. Persistent overlap induces superlinear coordination growth. Under irreducible coupling and bounded intrinsic regulation, such growth produces spectral instability. Stability under refinement therefore requires symmetry scaling in the form of hierarchical modular closure.

The result is conditional yet structural: refinement under finite relational constraint compels hierarchical organisation for persistence. Evolutionary stratification is thus interpreted as a geometric and dynamical necessity rather than a contingent historical artifact.

1 Relation as Environmental Projection

Evolution does not occur in an abstract combinatorial vacuum. All biological systems are embedded in an environment structured by physical law. Energy availability, thermodynamic gradients, chemical composition, spatial embedding, and information exchange define independent axes of survival constraint.

Although phenotypic diversity can expand combinatorially, the intrinsic dimensionality of environmental constraint space remains finite. No biological configuration escapes projection onto the same underlying survival coordinates.

Let d denote the intrinsic dimensionality of environmental constraint space. This d is fixed and independent of system size.

At refinement stage k, suppose N_k functional units exist. Each unit is associated with a normalised survival profile

$$u_i \in \mathbb{R}^d, \qquad \|u_i\| = 1.$$

These vectors encode projection onto independent environmental axes. They do not represent phenotype in full complexity, but the relational interface between organism and environment.

Relation arises because units share projection within the same finite space.

2 Coherence as Competitive Overlap

Competitive interaction strength between units is determined by shared projection:

$$C_{ij} = \langle u_i, u_j \rangle.$$

The matrix $G_k = (C_{ij})$ is the Gram matrix of the survival profiles. It is symmetric positive semidefinite and satisfies

$$\operatorname{rank}(G_k) \leq d.$$

Low rank reflects mediation through shared environmental axes, not weakness of coupling. All competitive interactions are filtered through the same finite-dimensional geometry.

The question is whether refinement can asymptotically eliminate overlap.

3 Geometric Constraint on Overlap

Because G_k has rank at most d, its eigenvalues $\lambda_1, \ldots, \lambda_{N_k}$ satisfy

$$\sum_{i=1}^{N_k} \lambda_i = \operatorname{tr}(G_k) = N_k,$$

and at most d are nonzero.

By Cauchy–Schwarz,

$$\sum_{i=1}^{N_k} \lambda_i^2 \geq \frac{\left(\sum_{i=1}^{N_k} \lambda_i\right)^2}{d} = \frac{N_k^2}{d}.$$

Since

$$\sum_{i=1}^{N_k} \lambda_i^2 = \|G_k\|_F^2 = \sum_{i,j} \langle u_i, u_j \rangle^2,$$

it follows that

$$\sum_{i \neq j} \langle u_i, u_j \rangle^2 \geq \frac{N_k^2}{d} - N_k.$$

Dividing by $N_k(N_k - 1)$ gives

$$\frac{1}{N_k(N_k-1)} \sum_{i \neq j} \langle u_i, u_j \rangle^2 \geq \frac{1}{d} - O\left(\frac{1}{N_k}\right).$$

Thus mean squared competitive overlap remains bounded below as refinement proceeds.

Refinement does not produce asymptotic orthogonality. Environmental dimensionality prevents it.

4 Positive-Density Coupling

Fix ϵ with $0 < \epsilon < 1/\sqrt{d}$ and define

$$\mathcal{P}_k(\epsilon) = \{(i,j) : i \neq j,\ |\langle u_i, u_j \rangle| \geq \epsilon\}.$$

A counting argument on the squared overlap sum yields

$$|\mathcal{P}_k(\epsilon)| \geq \frac{1/d - \epsilon^2}{1 - \epsilon^2} N_k^2$$

for sufficiently large N_k.

A positive fraction of all pairs therefore exhibit non-negligible competitive overlap.

Relation under finite constraint enforces coherence density.

5 Interpretive Consequence

As refinement increases N_k, units cannot distribute themselves into mutually orthogonal survival niches indefinitely. Environmental axes are reused. Overlap accumulates.

Competition is not accidental. It is geometrically forced.

The next stage of analysis examines how persistent coherence density reshapes system-level coordination and stability.

6 From Overlap to Coordination Structure

The previous section established that finite environmental dimensionality enforces a positive density of competitive overlap. A non-vanishing fraction of functional unit pairs remain non-negligibly coupled through shared projection onto survival axes.

This result is geometric. It does not yet address how overlap restructures the organisation of the system.

Biologically, competitive overlap is not inert. When two units share environmental projection, they cannot remain dynamically independent. They either suppress, regulate, compete, cooperate, specialise, or differentiate. Relation induces adjustment. Adjustment induces coordination.

Coherence is not static similarity; it is dynamic entanglement.

7 Definition of Effective Coordination Graph

Let ϵ be fixed as in the previous section and define the effective coordination graph

$$\Gamma_k = (V_k, E_k),$$

where $|V_k| = N_k$ and

$$(i,j) \in E_k \quad \text{if and only if} \quad |\langle u_i, u_j \rangle| \geq \epsilon.$$

$$|E_k| \geq c N_k^2,$$

with

$$c = \frac{1/d - \epsilon^2}{1 - \epsilon^2} > 0.$$

Thus total coordination density grows at least quadratically with system size.

8 Linear Growth of Coordination Degree

Define the coordination degree of unit i as

$$\deg_k(i) = |\{j \neq i : (i,j) \in E_k\}|.$$

Summing degrees yields

$$\sum_{i=1}^{N_k} \deg_k(i) = 2|E_k|.$$

Hence the mean coordination degree satisfies

$$\frac{1}{N_k} \sum_{i=1}^{N_k} \deg_k(i) = \frac{2|E_k|}{N_k} \geq 2cN_k.$$

Coordination degree grows linearly with refinement.

This is not a statistical artifact. It is a deterministic consequence of environmental geometry.

As system size doubles, mean coordination degree doubles.

9 Refinement as Cumulative Entanglement

Refinement corresponds to increasing N_k. Each new unit enters an already dense overlap structure. Because environmental axes are finite, new units cannot avoid projecting into previously occupied survival directions.

Thus each incremental unit contributes coordination edges proportional to existing system size.

Formally, if $N_{k+1} = N_k + 1$, then the new unit must satisfy

$$\deg_{k+1}(N_{k+1}) \geq cN_k,$$

for sufficiently large k.

Coordination growth is therefore cumulative. It scales with total prior refinement.

Superlinear coordination growth is not optional. It is forced by reuse of environmental axes.

10 Biological Interpretation

In biological evolution, increasing organismal diversity does not produce independence. It produces entanglement.

Species share energy pathways, chemical substrates, spatial embedding, and thermodynamic constraints. Cells share metabolic substrates. Proteins share regulatory pathways. Neural populations share sensory channels.

As refinement proceeds:

- Niche differentiation does not eliminate projection onto core survival axes.
- Resource partitioning reduces but does not nullify overlap.
- Innovation introduces variation within fixed constraint geometry.

Environmental dimensionality caps orthogonality.

Therefore coordination density cannot decay with system size.

11 Coherence Pressure

We now introduce a structural notion.

Coherence pressure refers to the growth of necessary coordination adjustments induced by overlap.

When mean coordination degree scales linearly in N_k, each unit must adjust relative to an increasing number of others.

Coherence pressure scales with system size.

This pressure does not yet imply instability. It implies increasing coordination burden.

The next section examines how such burden interacts with dynamical regulation.

12 Structural Summary

Finite-dimensional relation enforces persistent overlap. Persistent overlap induces dense coordination. Dense coordination forces linear growth of coordination degree. Total coordination growth becomes quadratic.

Coherence is therefore not diluted by refinement. It intensifies.

The question now becomes whether such intensification is dynamically sustainable.

13 From Coherence Pressure to Dynamical Amplification

Dense coordination alone does not imply instability. A system may exhibit extensive connectivity and remain stable if regulatory mechanisms scale appropriately.

The structural question is therefore precise:

When coordination density grows superlinearly under bounded environmental dimensionality, can intrinsic regulation scale sufficiently to preserve stability?

To answer this, we move from graph structure to dynamical systems.

14 Dynamical Framework

Let $x \in \mathbb{R}^{N_k}$ denote system state at refinement stage k. Assume the existence of a persistent configuration x^*.

Linearisation yields

$$\dot{x} = J_k x, \qquad J_k = -R_k + A_k,$$

where:

- R_k is diagonal and encodes intrinsic self-regulation.
- A_k encodes coordination effects between units.

Stability requires that all eigenvalues of J_k have negative real part.

15 Bounded Intrinsic Regulation

We assume intrinsic regulation is uniformly bounded:

There exists $\alpha > 0$ such that

$$0 \leq (R_k)_{ii} \leq \alpha \quad \text{for all } i, k.$$

This reflects a structural biological constraint: local stabilisation mechanisms do not scale proportionally with total system size. A cell, organism, or regulatory module does not double its intrinsic damping simply because global system size doubles.

Regulation is local. Coordination is relational.

16 Aggregate Coupling from Persistent Overlap

We have established that the effective coordination graph Γ_k contains at least

$$|E_k| \geq cN_k^2$$

edges for some constant $c > 0$ independent of k.

To translate structural overlap into dynamical interaction, we impose an aggregate non-degeneracy condition.

Aggregate Coupling Condition. There exists $c_0 > 0$ independent of k such that

$$\sum_{(i,j)\in E_k} (A_k)_{ij}^2 \geq c_0 N_k^2.$$

This condition does not assume uniform per-edge magnitude. It requires only that total interaction energy induced by persistent overlap scales proportionally with coordination density.

The assumption is minimal: if competitive overlap produces dynamically relevant interaction on a positive-density subset of pairs, the total squared magnitude of coupling must scale quadratically in N_k.

This assumption is independent of the geometric overlap result and represents a dynamical modelling condition.

17 Spectral Amplification under Dense Coordination

Consider the Frobenius norm:

$$\|A_k\|_F^2 = \sum_{i,j} (A_k)_{ij}^2.$$

Since the sum over E_k alone contributes at least $c_0 N_k^2$, we have

$$\|A_k\|_F^2 \geq c_0 N_k^2.$$

For any symmetric matrix,

$$\lambda_{\max}(A_k) \geq \frac{\|A_k\|_F}{\sqrt{N_k}}.$$

Hence

$$\lambda_{\max}(A_k) \geq \sqrt{c_0}\,\sqrt{N_k}.$$

The dominant eigenvalue grows at least on the order of $\sqrt{N_k}$.

This growth requires:

- no sign coherence,
- no specific structural alignment,
- no uniform per-edge lower bound.

It follows solely from dense magnitude accumulation under refinement.

18 Instability Threshold

Recall that

$$J_k = -R_k + A_k,$$

where intrinsic regulation satisfies

$$0 \leq (R_k)_{ii} \leq \alpha$$

for some fixed $\alpha > 0$ independent of k.

By Weyl's inequality,

$$\lambda_{\max}(J_k) \geq -\alpha + \lambda_{\max}(A_k).$$

Since $\lambda_{\max}(A_k) \geq \sqrt{c_0}\,\sqrt{N_k}$, there exists N^* such that for all $N_k > N^*$,

$$\lambda_{\max}(J_k) > 0.$$

Beyond this refinement threshold, perturbations grow exponentially. The system cannot maintain linear stability.

Instability therefore follows from:

- finite environmental dimensionality,
- persistent positive-density overlap,
- non-vanishing aggregate dynamical coupling,
- bounded intrinsic self-regulation,
- irreducible coordination structure.

19 Irreducibility and the Absence of Decomposition

The above amplification argument assumes that dense coupling contributes collectively.

If the coordination graph decomposes into independent modules of bounded size, then the Frobenius norm scales additively across blocks, and dominant eigenvalues remain bounded independently of total N_k.

Thus instability under superlinear coordination growth requires irreducibility: absence of bounded-size modular decomposition.

In irreducible dense systems, coherence pressure cannot be compartmentalised. Amplification accumulates globally.

Irreducible means: No partition of

$$V_k = \bigsqcup_{\ell=1}^{S_k} M_{k,\ell}$$

into subsets of uniformly bounded size such that all inter-subset coupling blocks vanish asymptotically.

20 Interpretive Meaning: Coherence Without Closure

Relation generates coherence. Coherence generates coordination density. Coordination density generates amplification.

Without structural closure into bounded modules, amplification scales faster than intrinsic regulation.

Coherence without closure destabilises.

21 Structural Summary

Superlinear coordination growth induces spectral amplification at least of order $\sqrt{N_k}$. Intrinsic regulation remains bounded. Irreducible refinement therefore produces dynamical instability.

Stability under refinement demands structural intervention.

The next section identifies the only structurally sufficient intervention mechanism.

22 Conditional Structural Necessity of Hierarchical Modularity

Theorem 1 (Conditional Structural Necessity)**.** *Let $\{N_k\}$ be a refinement sequence satisfying:*

1. *Environmental constraint space has fixed finite dimension d.*
2. *Functional units are represented by normalised survival profiles in $\mathbb{R}^d$.*
3. *Competitive overlap induces a coordination graph with positive-density edges: $|E_k| \geq cN_k^2$ for some $c > 0$.*
4. *Aggregate dynamical coupling satisfies*

$$\sum_{(i,j)\in E_k} (A_k)_{ij}^2 \geq c_0 N_k^2$$

   for some $c_0 > 0$ independent of k.
5. *Intrinsic regulation remains uniformly bounded: $0 \leq (R_k)_{ii} \leq \alpha$.*
6. *The coordination graph does not admit decomposition into modules of uniformly bounded size.*

Then there exists N^* such that for all $N_k > N^*$, the linearised system is spectrally unstable.

Conversely, stability under unbounded refinement requires decomposition into modules of uniformly bounded size, with the number of modules scaling linearly in N_k.

23 Closure as Dimensional Control

The previous section established that irreducible superlinear coordination induces spectral amplification that eventually overwhelms bounded intrinsic regulation.

Instability is not caused by connectivity alone. It arises from unbounded amplification of collective modes.

The structural question becomes:

What configuration prevents amplification from scaling with refinement?

The answer is closure through dimensional control of interaction rank.

24 Modular Decomposition

Suppose the coordination graph Γ_k admits a partition

$$V_k = \bigsqcup_{\ell=1}^{S_k} M_{k,\ell},$$

such that:

- Each module satisfies $|M_{k,\ell}| \leq B$ for some fixed B independent of k.
- Inter-module coupling magnitudes are uniformly bounded independently of N_k.

Under this structure, the interaction matrix A_k takes block form:

$$A_k = \begin{pmatrix} A_{k,1} & R_{12} & \cdots \\ R_{21} & A_{k,2} & \cdots \\ \vdots & \vdots & \ddots \end{pmatrix},$$

with each principal block dimension at most B.

25 Spectral Bound Under Modular Closure

Because each block has bounded size, its largest eigenvalue is bounded independently of total system size.

If inter-module coupling does not scale with N_k, then

$$\lambda_{\max}(A_k) \leq \max_{\ell} \lambda_{\max}(A_{k,\ell}) + \|R_k\|_2,$$

which remains bounded.

Thus bounded-size modular decomposition prevents the spectral growth that arises under irreducible dense coordination.

Amplification becomes compartmentalised.

26 Scaling Requirement

Since

$$N_k = \sum_{\ell=1}^{S_k} |M_{k,\ell}|,$$

and each module has size at most B, it follows that

$$S_k \geq \frac{N_k}{B}.$$

The number of stabilised modules must scale linearly with refinement.

Closure resources must scale proportionally with system size.

Hierarchy is therefore not ornamental. It is dimensional regulation.

27 Boundary Regimes

Two escape regimes exist:

Unbounded Environmental Dimensionality

If environmental dimensionality grows proportionally with refinement, competitive overlap may decay asymptotically, and superlinear coordination need not occur.

Vanishing Interaction Magnitude

If dynamical coupling strength decays sufficiently with system size, spectral amplification may remain bounded despite dense coordination.

Outside these regimes, hierarchical closure becomes structurally required.

28 Relation, Coherence, Closure

The argument can now be restated in structural language.

Finite-dimensional relation enforces persistent overlap. Persistent overlap generates coherence density. Coherence density induces superlinear coordination. Irreducible coordination produces amplification. Amplification without closure destabilises. Closure compartmentalises amplification. Closure modules must scale with refinement.

Relation induces coherence. Coherence demands closure. Closure manifests as hierarchy.

29 Exact Spectral Scaling Under Finite-Dimensional Geometric Projection

We now formalise the precise spectral growth law for the geometric overlap model introduced earlier.

Let $u_1, \ldots, u_N \in \mathbb{R}^d$ be normalised survival profiles with $\|u_i\| = 1$. Define the overlap matrix

$$C = UU^\top,$$

where $U \in \mathbb{R}^{N\times d}$ has rows u_i.

29.1 Reduction to a $d \times d$ Matrix

The nonzero eigenvalues of $C = UU^\top$ coincide with the eigenvalues of

$$M = U^\top U = \sum_{i=1}^{N} u_i u_i^\top \in \mathbb{R}^{d\times d}.$$

Thus

$$\lambda_{\max}(C) = \lambda_{\max}(M).$$

The problem reduces to analysing the spectral behaviour of M.

29.2 Expected Structure Under Isotropic Sampling

Assume the u_i are sampled independently and uniformly from the unit sphere S^{d-1}.

It is classical that

$$\mathbb{E}[u_i u_i^\top] = \frac{1}{d} I_d.$$

Therefore

$$\mathbb{E}[M] = \sum_{i=1}^{N} \mathbb{E}[u_i u_i^\top] = \frac{N}{d} I_d.$$

Hence the expected eigenvalues of M are all equal to N/d.

29.3 Concentration and Asymptotic Scaling

Each matrix $u_i u_i^\top$ is rank one and bounded in operator norm by 1. Standard matrix concentration results imply that with high probability,

$$\|M - \tfrac{N}{d} I_d\| \le O(\sqrt{N}).$$

Therefore

$$\lambda_{\max}(M) = \frac{N}{d} + O(\sqrt{N}).$$

We obtain the following result.

Proposition 2 (Linear Spectral Growth Under Finite Relational Rank)**.** *Let $u_1, \dots, u_N \in S^{d-1}$ be independent isotropic unit vectors in fixed dimension d. Define $C = UU^\top$.*

Then with high probability,

$$\lambda_{\max}(C) = \frac{N}{d} + O(\sqrt{N}),$$

and in particular,

$$\frac{\lambda_{\max}(C)}{N} \to \frac{1}{d} \quad \text{as } N \to \infty.$$

Thus in the geometric projection model, spectral amplification grows linearly in system size.

29.4 Interpretation

Under fixed environmental dimensionality d, each additional unit contributes expected mass $1/d$ along each constraint axis. These contributions accumulate coherently in the dominant eigen-direction.

The resulting leading eigenvalue scales as N/d.

Therefore instability thresholds scale linearly in system size under irreducible geometric coupling.

This is strictly stronger than the general $\sqrt{N}$ lower bound obtained earlier via Frobenius control.

29.5 Computational Validation

We validate the analytical scaling numerically by generating random unit profiles in fixed dimension and measuring $\lambda_{\max}(UU^\top)$.

For comparison, we also construct a truly modular system in which each block occupies a mutually orthogonal environmental subspace, eliminating cross-block overlap.

The following script was used to generate the reported numerical results.

```
import numpy as np
from numpy.linalg import eigvalsh

np.random.seed(42)

def generate_flat_profiles(N, d):
    U = np.random.normal(0, 1, (N, d))
    U /= np.linalg.norm(U, axis=1, keepdims=True)
    return U

def generate_true_modular_profiles(N, d, block_size):
    num_blocks = (N + block_size - 1) // block_size
    total_dim = num_blocks * d

    U = np.zeros((N, total_dim))

    for b in range(num_blocks):
        start = b * block_size
        end = min((b + 1) * block_size, N)
        dim_start = b * d
        dim_end = (b + 1) * d

        block = np.random.normal(0, 1, (end - start, d))
        block /= np.linalg.norm(block, axis=1, keepdims=True)

        U[start:end, dim_start:dim_end] = block
```

```
    return U

def overlap_matrix(U):
    return U @ U.T

alpha = 1.0
beta = 1.0
d = 5
block_size = 20

Ns = [50, 100, 200, 400, 800, 1600]

print("=== TRUE GEOMETRIC FLAT ARCHITECTURE ===")
for N in Ns:
    U = generate_flat_profiles(N, d)
    C = overlap_matrix(U)
    lam_A = eigvalsh(C)[-1]
    print(N, lam_A, lam_A/N)

print("\n=== TRUE BLOCK-ISOLATED MODULAR ARCHITECTURE ===")
for N in Ns:
    U = generate_true_modular_profiles(N, d, block_size)
    C = overlap_matrix(U)
    lam_A = eigvalsh(C)[-1]
    print(N, lam_A, lam_A/N)
```

In the flat geometric case, numerical results confirm

$$\lambda_{\max}(C) \approx \frac{N}{d}.$$

In the block-isolated modular case, $\lambda_{\max}(C)$ remains bounded independently of total system size, demonstrating spectral capping under hierarchical closure.

29.6 Dimensional Escape Regime

The amplification constant is inversely proportional to environmental dimension d.

Thus increasing relational dimensionality reduces spectral pressure.

This quantifies the dimensional escape regime: instability can be delayed only by increasing constraint rank or introducing modular closure.

30 Conclusion

Evolutionary refinement within finite-dimensional environments cannot indefinitely avoid increasing coordination density.

Superlinear coordination under bounded regulation generates instability unless interaction structure decomposes into bounded modules.

Hierarchical modular organisation therefore emerges as a conditional structural necessity for systems that refine under finite environmental constraint while maintaining stability.

This result does not depend on biological detail. It follows from geometry, coordination scaling, and spectral dynamics.

Evolutionary stratification is thus interpretable as a structural response to coherence pressure under constraint.